Nepal TST 2025 Solutions

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§1 Day 1 Problems

Problem 1.1. Consider a triangle $\triangle ABC$ and some point X on BC. The perpendicular from X to AB intersects the circumcircle of $\triangle AXC$ at P and the perpendicular from X to AC intersects the circumcircle of $\triangle AXB$ at Q. Show that the line PQ does not depend on the choice of X.

Problem 1.2. Find all integers n such that if

$$1 = d_1 < d_2 < \cdots < d_{k-1} < d_k = n$$

are the divisors of n, then the sequence

$$d_2 - d_1,\, d_3 - d_2,\, \ldots,\, d_k - d_{k-1}$$

forms a permutation of an arithmetic progression.

Problem 1.3. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(f(x)) + xf(xy) = x + f(y)$$

for all positive real numbers x and y.

§2 Day 2 Problems

Problem 2.1. Let the sequence $\{a_n\}_{n\geq 1}$ be defined by

$$a_1 = 1, \quad a_{n+1} = a_n + \frac{1}{\sqrt[2024]{a_n}} \quad \text{for } n \geq 1,\, n \in \mathbb{N}$$

Prove that

$$a_n^{2025} > n^{2024}$$

for all positive integers $n \geq 2$.

Problem 2.2. Kritesh manages traffic on a 45×45 grid consisting of 2025 unit squares. Within each unit square is a car, facing either up, down, left, or right. If the square in front of a car in the direction it is facing is empty, it can choose to move forward. Each car wishes to exit the 45×45 grid.

Kritesh realizes that it may not always be possible for all the cars to leave the grid. Therefore, before the process begins, he will remove k cars from the 45×45 grid in such a way that it becomes possible for all the remaining cars to eventually exit the grid.

What is the minimum value of k that guarantees that Kritesh's job is possible?

Problem 2.3. Consider an acute triangle ΔABC. Let D and E be the feet of the altitudes from A to BC and from B to AC respectively.

Define D_1 and D_2 as the reflections of D across lines AB and AC, respectively. Let Γ be the circumcircle of ΔAD_1D_2. Denote by P the second intersection of line D_1B with Γ, and by Q the intersection of ray EB with Γ.

If O is the circumcenter of ΔABC, prove that O, D, and Q are collinear if and only if quadrilateral $BCQP$ can be inscribed within a circle.

§3 Solutions to Day 1

§3.1 Problem 1, proposed by Shining Sun(USA)

Problem 3.1. Consider a triangle $\triangle ABC$ and some point X on BC. The perpendicular from X to AB intersects the circumcircle of $\triangle AXC$ at P and the perpendicular from X to AC intersects the circumcircle of $\triangle AXB$ at Q. Show that the line PQ does not depend on the choice of X.

Proof. Denote by O the circumcenter of $\triangle ABC$. We show that PQ always lies on the line AO, equivalent to showing $A - O - P$ and $A - O - Q$. Let D be the point on AC such that $XD \perp AC$.

Claim 3.2 — We have $A - O - Q$.

Proof.

$$\measuredangle BAO = \frac{180^\circ - \measuredangle AOB}{2} = 90^\circ - \measuredangle ACB = \measuredangle CXD = \measuredangle BAQ$$

□

Claim 3.3 — We have $A - O - P$.

Proof.

$$\measuredangle CAP = \measuredangle CXP = 90^\circ + \measuredangle CBA = \measuredangle CAO$$

□

Therefore, P and Q lie on AO, we're done. □

§3.2 Problem 2, proposed by Kritesh Dhakal(Nepal)

Problem 3.4. Find all integers n such that if

$$1 = d_1 < d_2 < \cdots < d_{k-1} < d_k = n$$

are the divisors of n, then the sequence

$$d_2 - d_1,\ d_3 - d_2,\ \ldots,\ d_k - d_{k-1}$$

forms a permutation of an arithmetic progression.

Proof. □

§3.3 Problem 3, proposed by Andrew Brahms(USA)

Problem 3.5. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(f(x)) + xf(xy) = x + f(y)$$

for all positive real numbers x and y.

Proof. Let $P(x, y)$ be the given assertion. We claim that the only solutions for f are $f(x) = 1$ and $f(x) = \frac{c}{x}$ for some constant c. These work upon checking.

When f is constant, we get $f(x) = 1$, so from henceforth assume that f is not constant.

$$P(1, x) \implies f(f(1)) + f(x) = 1 + f(x) \implies f(f(1)) = 1$$

$$P(x, f(x)) \implies f(f(x)) + xf(xf(x)) = x + f(f(x)) \implies f(xf(x)) = 1$$

Therefore, if f was injective, we would have $xf(x) = f(1)$ and we would be done. So, we prove the following claim.

Claim 3.6 — f is injective.

Proof. Assume that $f(a) = f(b)$ for some a, $b \in \mathbb{R}^+$. We note:

$$P(x, a) - P(x, b) \implies f(ax) = f(bx)$$

$$P(a, x) \implies f(f(a)) + af(ax) = a + f(x)$$

$$P(b, x) \implies f(f(b)) + bf(bx) = b + f(x)$$

Combining gives us $a(1 - f(ax)) = b(1 - f(bx))$, and since f isn't constant, we can conclude $a = b$. □

And we're done, as from $P(x, f(x))$ as seen above we get $xf(x) = f(1) \implies f(x) = \frac{c}{x}$, setting $c = f(1)$. □

§4 Solutions to Day 2

§4.1 Problem 4, proposed by Prajit Adhikari(Nepal)

Problem 4.1. Let the sequence $\{a_n\}_{n\geq 1}$ be defined by

$$a_1 = 1, \quad a_{n+1} = a_n + \frac{1}{\sqrt[2024]{a_n}} \quad \text{for } n \geq 1,\, n \in \mathbb{N}$$

Prove that

$$a_n^{2025} > n^{2024}$$

for all positive integers $n \geq 2$.

> **Remark 4.2.** In my opinion, this is the absolute easiest problem of the six. This problem is easily solved using induction, as we will see below. I've also heard/seen about solutions using calculus, but I think most of them are inductive too, so it seems induction is the way to go for this problem.

Proof. Note that for all k, $a_{k+1} > a_k$. We proceed via induction.

The base case $n = 2$ is obvious as $a_2^{2025} = 2^{2025} > 2^{2024}$ holds. Assume now, for some k that $a_k^{2025} > n^{2024}$. It suffices to show that $a_{k+1}^{2025} > n^{2024}$ or equivalently, $a_{k+1}^{\frac{2025}{2024}} > k+1$. This is easy, however, as:

$$\begin{aligned}
a_{k+1}^{\frac{2025}{2024}} &= a_{k+1} \cdot a_{k+1}^{\frac{1}{2024}} \\
&> a_{k+1} \cdot a_k^{\frac{1}{2024}} \\
&= \left(a_k + \frac{1}{a_k^{\frac{1}{2024}}} \right) a_k^{\frac{1}{2024}} \\
&= a_k^{\frac{2025}{2024}} + 1 \\
&> k+1
\end{aligned}$$

And we're done. $\square$

§4.2 Problem 5, proposed by Shining Sun(USA)

Problem 4.3. Kritesh manages traffic on a 45×45 grid consisting of 2025 unit squares. Within each unit square is a car, facing either up, down, left, or right. If the square in front of a car in the direction it is facing is empty, it can choose to move forward. Each car wishes to exit the 45×45 grid.

Kritesh realizes that it may not always be possible for all the cars to leave the grid. Therefore, before the process begins, he will remove k cars from the 45×45 grid in such a way that it becomes possible for all the remaining cars to eventually exit the grid.

What is the minimum value of k that guarantees that Kritesh's job is possible?

Proof. Define two cars in the same row/column facing opposite directions to be in an **opposition**. Obviously the only way to have one of the cars to escape the **opposition** is to remove one of them. Further, Car A is said to **block** Car B if it is in the same row(if Car B faces left/right) or column(if Car B faces up/down) but the two cars are not in an **opposition**.

We claim the answer is 1012. First, we show that this answer is achievable.

The construction consists of all the cars in the first 22 columns facing left, then columns 23-45 facing right, and 22 cars facing up in the middle row with the remaining facing down. Then, obviously the minimum required amount of cars to be removed is $22 \times 45 + 22 = 1012$.

We now show that this is the minimum required to clear the board. Consider the 45×45 grid, then WLOG let the number of cars facing right be less than the number of cars facing left, and the number of cars facing up be less than the number of cars facing down. Then, to remove all of the **oppositions** on the board, we remove all of the cars facing right and facing up. This way, at most $\left\lfloor \frac{2025}{2} \right\rfloor$ cars are removed. With all of the cars remaining facing left and down, we are left to show that this can clear the board. We notice that the cars can only **block** each other. Consider the bottommost row, then remove all the cars facing down, this way there are no down cars **blocking** the left cars, and so the bottommost row is cleared, we can remove it from the grid. Repeat this similarly for the second bottommost row, and we can continue this until the final row, clearing the grid as required. $\square$

> **Remark 4.4.** I apologize for the absolute horror above. Maybe one day I'll include asy diagrams and make it easier to read.

§4.3 Problem 6, proposed by Kritesh Dhakal(Nepal)

Problem 4.5. Consider an acute triangle ΔABC. Let D and E be the feet of the altitudes from A to BC and from B to AC respectively.

Define D_1 and D_2 as the reflections of D across lines AB and AC, respectively. Let Γ be the circumcircle of ΔAD_1D_2. Denote by P the second intersection of line D_1B with Γ, and by Q the intersection of ray EB with Γ.

If O is the circumcenter of ΔABC, prove that O, D, and Q are collinear if and only if quadrilateral $BCQP$ can be inscribed within a circle.

Proof. □